Worksheet 5-6 A Concavity and Points of Inflection #1
AP Calculus Name_____

Directions: For each of the following, identify the intervals of concavity for $f(x)$ and find any value(s) of x where $f(x)$ has a point of inflection. DO ALL WORK ON NOTEBOOK PAPER

1. $f''(x) = 5(x-4)^2(x+3)$

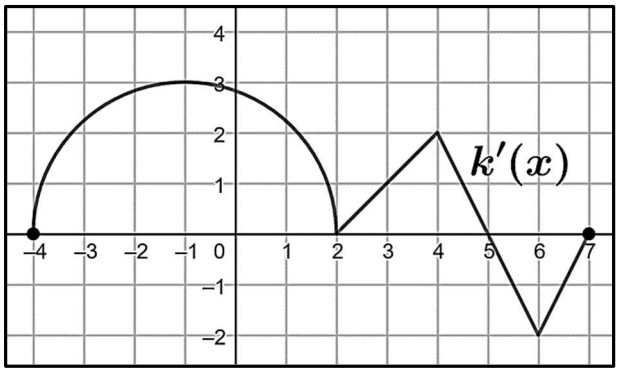
2. $f(x) = x^4 - 6x^2$

3. $f'(x) = x^2 - 3x - 10$

4. $f'(x) = -x^3 - 6x^2 + 63x - 2$

5. $f'(x) = e^{x^2-4x}$

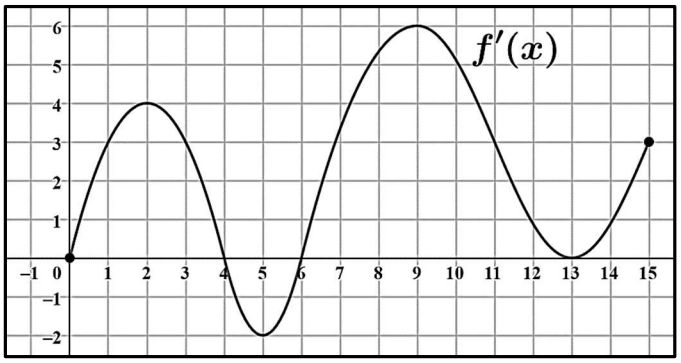
6. $f(x) = \dfrac{x}{x-3}$



7. The function k is continuous and differentiable on the interval $[-4, 7]$. The graph of k', the derivative of k, consists of three line segments and a semicircle as shown in the figure above.

A) On what open interval(s) is $k(x)$ concave down? Give a reason for your answer.

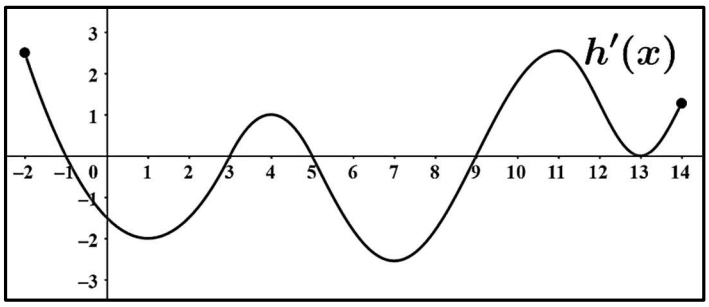
B) For which value(s) of x does $k(x)$ have a point of inflection? Give a reason for your answer.



8. The function f is continuous and twice differentiable on the interval $[0, 15]$. The graph of f', the derivative of f, is given above and has horizontal tangents at $x = 2,\ 5,\ 9,$ and 13.

A) On what interval(s) is $f(x)$ concave down? Explain your reasoning.

B) For which value(s) of x does $f(x)$ have a point of inflection? Give a reason for your answer.



9. The function h is continuous and twice differentiable on the interval $[-2, 14]$. The graph of h', the derivative of h, is given above and has horizontal tangents at $x = 1,\ 4,\ 7,\ 11,$ and 13.

A) For which value(s) of x does $h(x)$ have a relative minimum? Give a reason for your answer.

B) For which value(s) of x does $h(x)$ have a point of inflection? Give a reason for your answer.

C) On what interval(s) is the graph of $h(x)$ both decreasing and concave down? Explain your reasoning.